



CLUSTER INNOVATION CENTRE · CIC-TBI · PROGRESS REVIEW

Quarterly Progress Report

July to September 2026, month by month

TO	Mr Pratik Kumar, Project Manager, CIC-TBI, Cluster Innovation Centre, University of Delhi
FROM	Protopus (Soham, Founder)
SUBMITTED	5 October 2026
FOR THE MEETING	12 October 2026, 2:00 PM, Director's Office, 3rd Floor, Cluster Innovation Centre, University Stadium, Gate No. 2
PERIOD COVERED	1 July to 30 September 2026
LINKS	protopus-kappa.vercel.app (website) · probe-matrix.vercel.app (Probe Matrix concept)

Dates are approximate (early, mid or late in the month). Prices and costs are projections from our cost report and are not yet validated by pilots.

SUMMARY OF THE QUARTER

From a researched idea to a prototype under build, with funding conversations open.

Protopus is building smart machines and accessories for hardware prototyping, guided by an AI engineer called Proto. [Baker](#), the first machine, makes a finished circuit board on the desk. [Mixer](#), the second, tests it. Over the quarter we completed our research and website (July), finalised the prototype design and started building it (August), and integrated software with hardware while opening funding talks for the proof of concept (September). We also conceptualised a new accessory, the [Probe Matrix](#).

Jul	Aug	Sep
research, website, investor meeting	design finalised, build started	integration, funding talks, Probe Matrix

STATUS AT A GLANCE · 30 SEPTEMBER 2026

MILESTONE	MONTH	STATUS
Market and competition research	Jul	DONE
Website live (Baker and Mixer pages)	Jul	DONE
Meeting with Samuel Beek (Schematik): funding offered after the proof of concept	Jul	DONE
Prototype design finalised	Aug	DONE
Build of the first Baker prototype	Aug-	IN PROGRESS
Software-hardware integration	Sep-	IN PROGRESS
Proof-of-concept funding from STMicroelectronics	Sep-	IN DISCUSSION
Probe Matrix accessory	Sep	CONCEPT
Costing, workflow, China manufacturing plan and pitch documents	Sep	DONE



Approximate positions within each month.

THE QUARTER IN NUMBERS

What the three months produced.

15

[Baker](#) process steps designed, from bare copper board to reflowed board

70

line items in the costed [Baker](#) bill of materials

2 + 1

machines architected ([Baker](#), [Mixer](#)) and one accessory conceived ([Probe Matrix](#))

₹3,73,418

costed build of the first [Baker](#) prototype, incl. duty and 15% contingency

₹1,44,842

projected [Baker](#) production cost per unit at 100 units

7

deliverables completed: website, BOM, cost report, workflow, China plan, pitch deck, concept page

DELIVERABLES COMPLETED THIS QUARTER

DELIVERABLE	WHAT IT CONTAINS	MONTH
Protopus website	Product site with Baker and Mixer pages	Jul
Prototype design	Baker 15-step process, machine layout, subsystem specification; Mixer architecture	Aug
Bill of materials and costing workbook	Baker and Mixer : prototype BOM, cost breakup, China-parts production BOM and unit cost	Sep
Cost report	Prototype and production cost for Baker and Mixer , with duty and contingency	Sep
Workflow document	"The story of a PCB": Baker 's 15 steps and Mixer 's test flow, step by step	Sep
China manufacturing plan	Turnkey contract manufacturing in China, with testing and minimal assembly in India	Sep
Pitch deck	Problem, solution, competition, economics, Baker -first plan and funding ask	Sep
Probe Matrix concept page	probe-matrix.vercel.app	Sep

Research, brand and website, and our first investor conversation.

July was about understanding the problem, the people who face it and the market, and then putting Protopus in front of the world and of investors.

1.1 Problem study: how a prototype is made today

EARLY JULY · DONE

We mapped the full journey of a first circuit board in India. It needs a fab order (JLCPCB or PCBWay), 7–20 days of shipping and customs, hand soldering or a paid assembly service, and separate instruments to test it. Each try of an assembled board from JLCPCB, landed in India, costs about ₹6,700. People lose weeks per revision, and many stop.

1.2 User study: where people get stuck

EARLY JULY · DONE

We broke hardware development into eight rungs: idea to specification, parts, schematic, layout, fabrication, assembly, bring-up and iteration. Each one needs new knowledge, and each one loses people. Rungs 1–4 are already being addressed by AI design software. Rungs 5–8 are physical, which is where we decided to work.

1.3 Competition study

EARLY TO MID JULY · DONE

We studied the tools that exist and their prices: desktop PCB mills (LPKF ProtoMat, Makera Carvera), the Bantam Tools mill (discontinued in 2025), Voltera's ink printer, Neoden pick-and-place machines, USB instruments such as the Digilent Analog Discovery 3, fabs, and AI design tools such as Flux.ai. Each solves only one step. None runs the whole loop from design to tested board, and none is driven by an AI.

1.4 Market study

MID JULY · DONE

We sized our first market in India: 5,868 AICTE engineering colleges, about 2.25 lakh DPIIT-recognised startups (of which we assume about 1% build hardware), and 10,000 Atal Tinkering Labs, with 50,000 more announced. We also placed this within the wider PCB and test-and-measurement markets.

1.5 Product concept and positioning

MID JULY · DONE

We defined the system. Proto is an AI engineer that plans the build. [Baker](#) is a machine that makes the board: mill, drill, solder mask, stencil, paste and reflow. [Mixer](#) is an instrument that tests it. We positioned the machines for prototyping, the first 1-10 boards of a design, while volume stays with fabs.

1.6 Brand identity and website launch

MID TO LATE JULY · DONE

We created the Protopus identity: name, logo, the platypus mascot and our design language. We then designed and published the website, with dedicated pages for [Baker](#) and [Mixer](#): protopus-kappa.vercel.app.

1.7 Meeting with Samuel Beek, Founder and CEO of Schematik

LATE JULY · DONE

We presented Protopus to Samuel Beek. He committed to fund Protopus once the proof of concept is achieved. This is a verbal commitment, and it depends on the proof of concept.

Prototype design finalised; the real prototype started.

In August we turned the concept into an engineering design, chose every part, and began building the first [Baker](#).

2.1 [Baker](#) process design: 15 steps

EARLY AUGUST · **DONE**

We fixed the 15-step sequence that turns a bare copper board into a finished one. In order, the machine feeds the blank and probes its surface; mills the traces and drills the holes; cleans the board; coats, UV-images, develops and cures the solder mask; laser-cuts the stencil and applies solder paste; places the parts using a printed jig; reflows the board; and hands it over for test.

2.2 Machine architecture

EARLY TO MID AUGUST · **DONE**

We chose one moving shuttle that carries the board through two process zones in a single coordinate frame. Zone A mills, drills and laser-cuts the stencil. Zone B coats, images and cures the solder mask. A refill magazine feeds the next blank, so a board does not need loading each time. The placement jig is generated from the Gerber and printed inside the machine.

2.3 Subsystem specification

MID AUGUST · **DONE**

Zone A: a 500 W ER11 spindle with a touch probe, and a 10 W, 445 nm laser for the stencil. Zone B: a syringe-dispensed liquid solder mask, imaged by a 10.1-inch, 405 nm LCD. Reflow: a 300 W silicone hotplate with SSR control and a MAX31855 thermocouple, using low-temperature Sn42Bi58 solder (138 °C). Control: a Raspberry Pi 5 with a BTT Octopus Pro board, running Klipper.

2.4 [Mixer](#) architecture (for a later stage)

MID AUGUST · **DONE**

We defined [Mixer](#) as eight instruments on one cable: a 2-channel oscilloscope, a 2-channel waveform generator, a 16-channel logic analyser, a pattern generator, a protocol decoder, two programmable supplies and a 24-bit multimeter. It is built around a Tang Primer 20K FPGA. [Mixer](#) is planned for after the [Baker](#) proof of concept.

2.5 Parts selection and bill of materials

MID TO LATE AUGUST · DONE

We selected the parts for every [Baker](#) subsystem (motion, spindle and laser, mask and UV, reflow, control electronics, frame) and built the bill of materials, so that procurement could start.

2.6 Build of the real prototype started

LATE AUGUST · IN PROGRESS

We started procurement and assembly of the first [Baker](#) prototype.

Integration, proof-of-concept funding, a new accessory, and the business plan.

In September we connected the software to the machine, opened funding talks for the proof of concept, conceptualised the [Probe Matrix](#), and documented the costs, workflow and manufacturing plan.

3.1 [Baker](#) Studio and Proto: the software layer

THROUGH SEPTEMBER · **IN PROGRESS**

We worked on [Baker](#) Studio, the software in which Proto checks a design and plans the machine job. Proto reads schematics and layouts through the Flux.ai API, flags problems such as traces that are too thin, proposes a fix for the user to accept, and plans the mill, drill, mask, stencil and jig. Proto's character is built on Gemini.

3.2 Software-hardware integration

THROUGH SEPTEMBER · **IN PROGRESS**

We worked on connecting [Baker](#) Studio to the prototype's control electronics through the Klipper firmware, with Moonraker and a Proto job runner, so that a planned job can drive the machine.

3.3 Costed bill of materials and cost report

MID TO LATE SEPTEMBER · **DONE**

We costed the [Baker](#) bill of materials line by line (70 items). The first [Baker](#) prototype comes to ₹3,73,418, including import duty and courier on imported parts and a 15% contingency. The [Mixer](#) prototype comes to ₹74,847. With parts sourced from China and assembly in Delhi at 100 units, production cost is projected at ₹1,44,842 for [Baker](#) and ₹13,740 for [Mixer](#).

3.4 Workflow document: "the story of a PCB"

LATE SEPTEMBER · **DONE**

We documented [Baker](#)'s 15 steps, from a bare copper board to a reflowed board, followed by [Mixer](#)'s test flow and its verdict, as one continuous story.

3.5 China manufacturing plan

LATE SEPTEMBER · **DONE**

We planned turnkey contract manufacturing in China, with testing, minimal assembly and shipping from India. The plan covers supplier selection, product-introduction gates, line stations, the test system, quality, landed cost, IP, logistics and compliance.

3.6 Raising proof-of-concept funding from STMicroelectronics

MID TO LATE SEPTEMBER · **IN DISCUSSION**

We approached STMicroelectronics to fund the prototype and the proof of concept. Discussions are ongoing; nothing has been committed yet.

3.7 [Probe Matrix](#) conceptualised

LATE SEPTEMBER, AFTER THE STMICROELECTRONICS MEETING · **CONCEPT**

Following that meeting, we conceptualised the [Probe Matrix](#). It is an accessory that places spring probes under the board, so Proto can reach every net to test and program the board without a person moving probes. We published a concept page: probe-matrix.vercel.app.

3.8 Pitch deck and round-1 review

LATE SEPTEMBER · **DONE**

We prepared the pitch deck (problem, solution, competition, economics, plan and funding ask) and received round-1 review questions, for which we have prepared written answers. As a result, the plan is now [Baker](#) first, with [Mixer](#) and the [Probe Matrix](#) after the proof of concept and funding released in milestone-based instalments.

What we will do next, one milestone at a time.

The next quarter is about finishing the first [Baker](#) prototype and closing the funding for the proof of concept. [Mixer](#) and the [Probe Matrix](#) stay at design stage until the proof of concept is achieved.

PLANNED STEP	TARGET
Complete the first Baker prototype build	October–November
Run all 15 Baker steps end to end on one board	November–December
Continue software–hardware integration; Proto from a plain-English brief to Gerbers to a Baker job	November–December
Close proof-of-concept funding (STMicroelectronics and others)	October–December

FUNDING PLAN

We are seeking ₹15 lakh for a nine-month proof of concept, or ₹10 lakh in a lean version. We propose it is released in three instalments, each only after the previous milestone is shown: ₹6 lakh at signing; ₹5 lakh once [Baker](#) runs all 15 steps on one board; ₹4 lakh once the process tests pass and Proto produces a [Baker](#) job from a brief. The proof of concept is 20 boards made with Proto + [Baker](#). No founder salary is drawn.

LINKS

WEBSITE	https://protopus-kappa.vercel.app/
BAKER	https://protopus-kappa.vercel.app/#/baker
MIXER	https://protopus-kappa.vercel.app/#/mixer
PROBE MATRIX	https://probe-matrix.vercel.app/

Submitted by Soham, Founder, Protopus · soham9763015@ece.du.ac.in